Online Retail Sales & Operations Analysis

Project brief | PostgreSQL + Power BI | Portfolio project

Business problem

Retail teams need a clear view of sales performance, product contribution, country performance, customer purchasing behavior, and unusual transaction patterns. Raw transaction exports can contain missing customer identifiers, blank descriptions, cancellation-prefixed invoices, negative quantities, and zero prices.

Project objective

Use PostgreSQL to audit and prepare the data, answer business questions with documented SQL, and design a Power BI dashboard that supports commercial decisions and operational review.

Dataset and scope

The project uses the UCI Online Retail dataset, containing historical transactions from a UK-based online retailer. The current local import contains 541,909 rows. The exact date range and date parsing should be validated against the imported data before final publication.

Method

1. Load a raw staging table with text fields. 2. Profile missingness and unusual transaction patterns. 3. Create a clean analytical view with numeric quantity/price and normalized blank identifiers. 4. Answer business questions with SQL. 5. Build a three-page Power BI report. 6. Document actual findings, limitations, and recommendations.

Core definitions

Gross positive line-item revenue = quantity × unit price for rows where quantity > 0 and unit_price > 0. This is a baseline sales measure, not net revenue after returns. C-prefixed invoice numbers are treated as a useful indicator for investigation, not absolute proof of business meaning.

Deliverables

SQL business question bank; Power BI dashboard; project README; project brief; analysis report with actual findings to be completed; data-quality notes; and reproducible instructions.

Limitations

Customer IDs are missing on many rows. Negative quantities can represent returns, corrections, damaged goods, or other adjustments. Do not silently delete records or claim net revenue without validated return/cancellation treatment. Do not publish the raw CSV unless dataset redistribution terms permit it.